

MongoDB: The Most Popular NoSQL Database

Here are the basics of MongoDB, the database used by thousands of organisations, including 30 of the top 100 companies in the world. Many of the world's most innovative Web companies use MongoDB.



MongoDB, considered as the most popular NoSQL database, is an open source document database that provides high performance, high availability and automatic scaling.

It has the following key features.

High performance: MongoDB provides high performance data persistence. In particular, its support for embedded data models reduces I/O activity on database systems. Its indexes support faster queries and can include keys from embedded documents and arrays.

High availability: To ensure high availability, MongoDB has the replication facility (also called replica sets), which provides the following:

- Automatic failover
- Data redundancy

A replica set is a group of MongoDB servers that maintain the same data, providing redundancy and increasing data availability.

Automatic scaling: MongoDB provides horizontal scalability as part of its core functionality. Automatic sharding distributes data across a cluster of machines. Replica sets can provide eventually-consistent reads for low-latency high throughput deployments.

Installing MongoDB

MongoDB can be installed on various OSs in the following manner.

Red Hat

- Create a `/etc/yum.repos.d/mongodb.repo` file to hold the following configuration information for the MongoDB repository:

```
[mongodb]
name=MongoDB Repository
baseurl=http://downloads-distro.mongodb.org/repo/redhat/os/
x86_64/
gpgcheck=0
enabled=1
```

- Install the MongoDB packages and associated tools as follows:

```
sudo yum install mongodb-org
```

Ubuntu

- Import the public key used by the package management system:

```
sudo apt-key adv --keyserver hkp://keyserver.ubuntu.com:80
--recv 7F0CEB10
```

Create the `/etc/apt/sources.list.d/mongodb.list` list file using the following command:

```
echo 'deb http://downloads-distro.mongodb.org/repo/ubuntu-
```